THE CENTRAL BANK'S INSTITUTIONAL MEMORY

Knowledge Graphs, Temporal Precedents and Governed Institutional Judgment

Alejandro Reynoso | October 2026

The learning question

What would it mean for a central bank to remember?

A decision is not an isolated document. It is a historical relationship among **circumstances, evidence, judgment, action, and outcome**.

- Can we reconstruct what policymakers knew *at the time*?
- Can we identify valid precedents without mistaking textual similarity for causal relevance?
- Can an LLM improve interpretation while preserving human institutional authority?

The policy problem: a synthetic liquidity shock

Observed symptoms

- Repo rates spike
- Collateral haircuts widen
- Solvent dealers face funding pressure
- Market functioning deteriorates

Committee decision

- Targeted collateralized support?
- Monitoring and communication?
- Operational intervention?
- Wait for counterfactual evidence?

Teaching simulation only: no actual central-bank data or policy recommendation.

The conceptual architecture

Persistent vault → typed graphs → governed paths

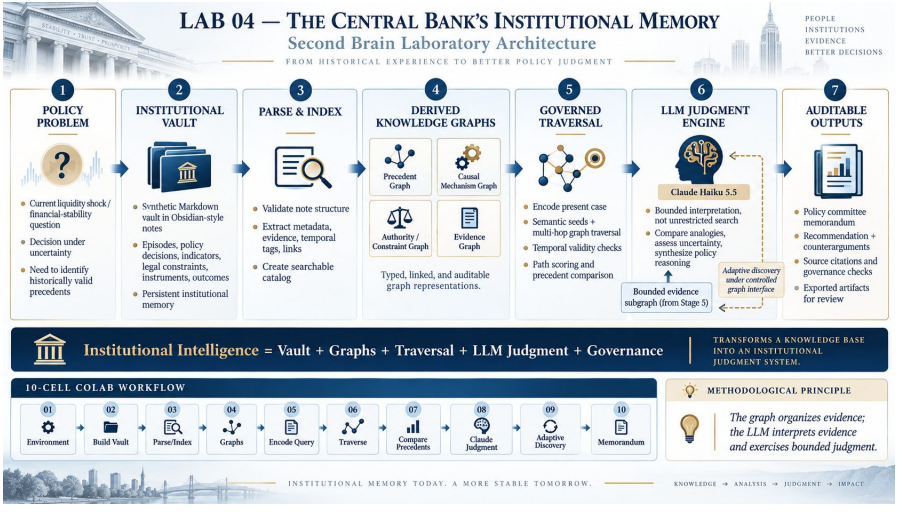
→ precedent comparison → bounded LLM judgment

→ auditable memorandum

Design principle

The graph structures and filters evidence; the model interprets it. Governance binds both processes.

General laboratory architecture



Ten-code-cell pedagogical sequence

1. Environment and persistence
2. Synthetic institutional vault
3. Parse, validate, index
4. Typed knowledge graph
5. Encode question and seeds
6. Governed multi-hop traversal
7. Compare historical precedents
8. Claude Haiku 5.5 judgment
9. Bounded adaptive discovery
10. Memo, evaluation and audit

Pedagogical contrast

Deterministic retrieval builds an inspectable evidence backbone; model judgment and controlled discovery operate on top of it.

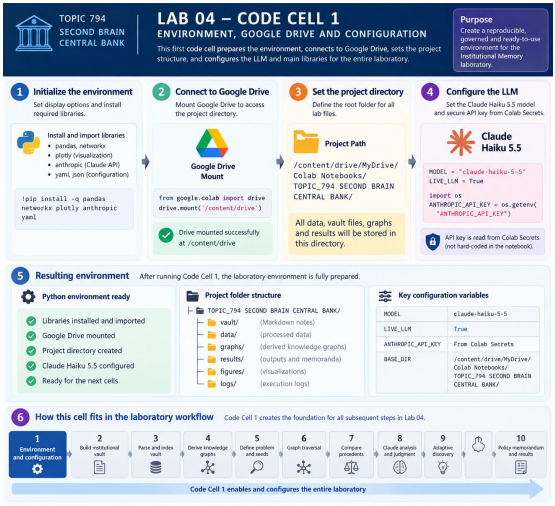
Code Cell 01 | Environment and persistent configuration

METHOD AND INSTITUTIONAL DESIGN

- Mount Google Drive, define the project root, import dependencies, establish reproducibility and secure Claude credentials.
- `/content/drive/MyDrive/Colab Notebooks/TOPIC_794 SECOND BRAIN CENTRAL BANK`

Key lesson The operating environment persists vault data, graph exports, figures, results and execution logs.

Code Cell 01 | instructional infographic



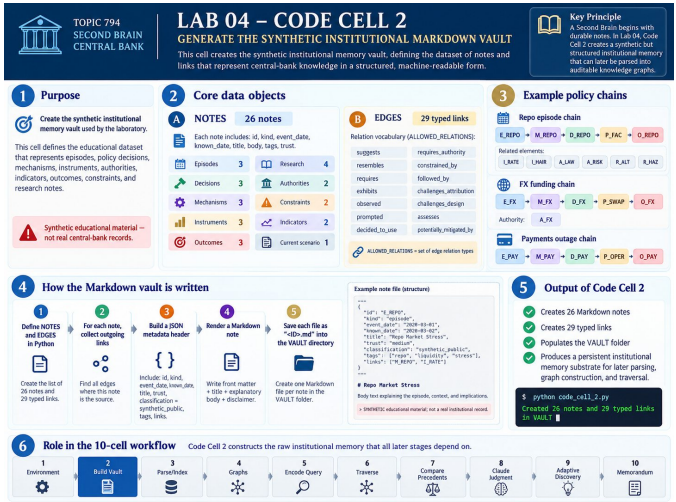
Code Cell 02 | Constructing the institutional vault

METHOD AND INSTITUTIONAL DESIGN

- Create 26 synthetic Markdown notes and 29 directed, typed links across episodes, decisions, mechanisms, instruments and outcomes.
- Each note records: ID, kind, event date, known date, title, trust, tags, outgoing links and provenance.

Key lesson Markdown is the durable source of truth; the derived graph is a computational view.

Code Cell 02 | instructional infographic



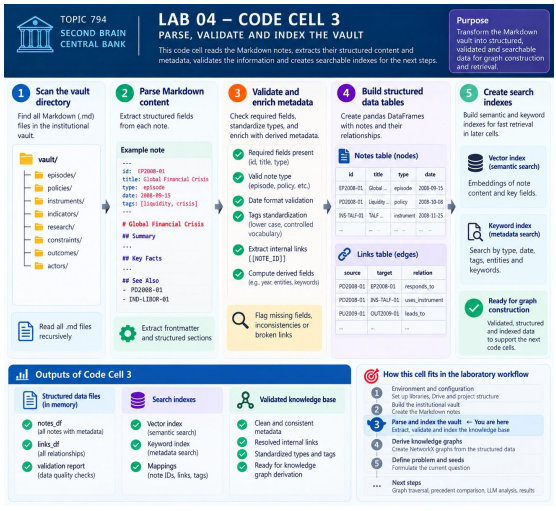
Code Cell 03 | Parsing, validation and indexing

METHOD AND INSTITUTIONAL DESIGN

- Parse JSON metadata from Markdown front matter, validate references and create a catalog indexed by stable identifiers.
- Reject or flag malformed notes, unresolved IDs and invalid relationships before graph construction.

Key lesson A validated vault prevents later traversal from silently inventing evidence.

Code Cell 03 | instructional infographic



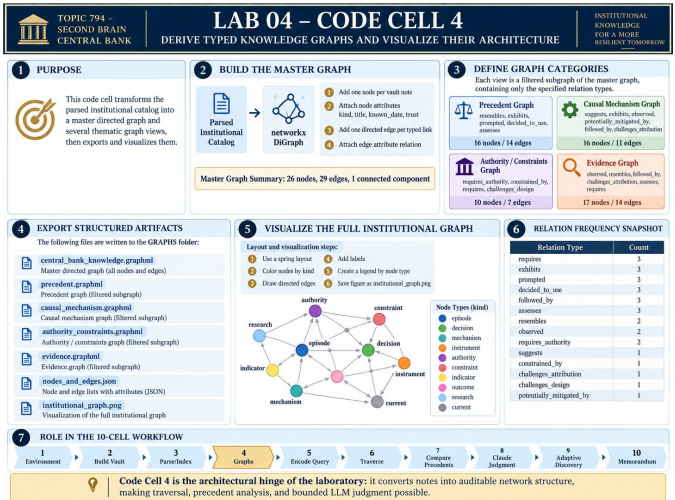
Code Cell 04 | Typed knowledge graphs

METHOD AND INSTITUTIONAL DESIGN

- Build a master NetworkX directed graph and four typed subgraph views; export GraphML and JSON.
- Master: 26 nodes / 29 edges. Precedent: 16/14; causal: 16/11; authority: 10/7; evidence: 17/14.

Key lesson A single institutional knowledge base supports several distinct analytical questions.

Code Cell 04 | instructional infographic



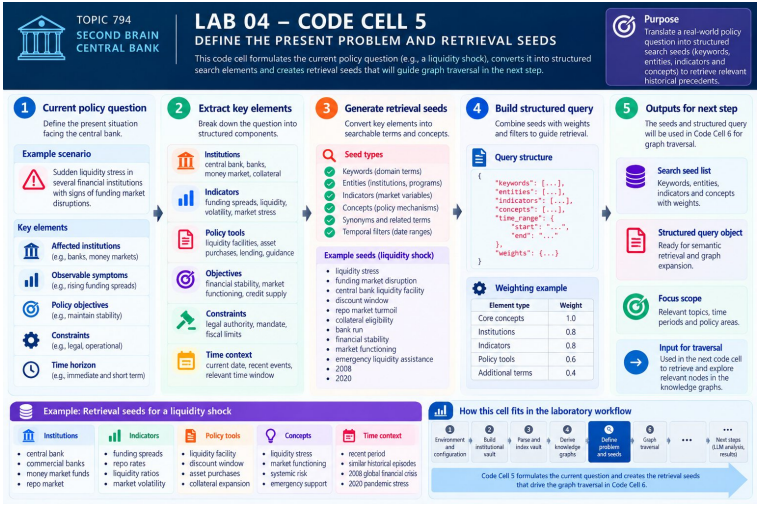
Code Cell 05 | Problem specification and retrieval seeds

METHOD AND INSTITUTIONAL DESIGN

- Represent the current scenario as a query and produce traceable starting nodes for subsequent traversal.
- Keyword overlap and cutoff-aware ranking lead to specified seeds: C_CURRENT, M_REPO, E_REPO, I_HAIR.

Key lesson This is grounded deterministic seed selection, not an LLM-designed search strategy.

Code Cell 05 | instructional infographic



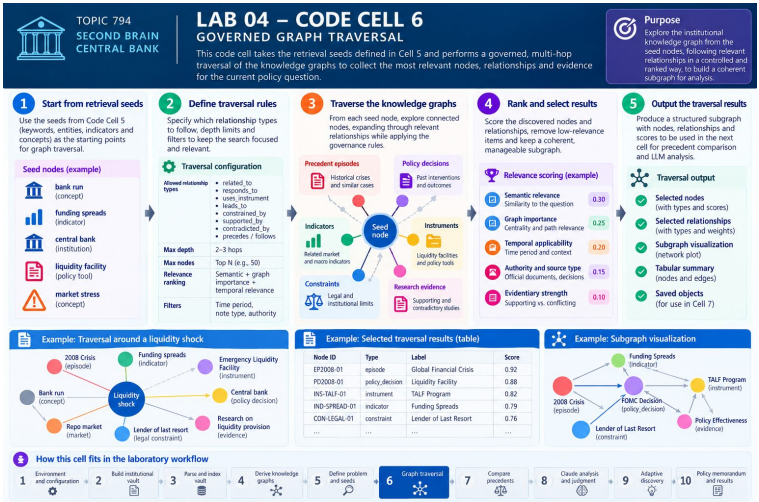
Code Cell 06 | Governed multi-hop traversal

METHOD AND INSTITUTIONAL DESIGN

- Expand from seeds along the directed evidence network while preserving paths and node provenance.
- Limits include clearance=synthetic_public, maximum four hops, at most 26 candidates, plus time/authority checks.

Key lesson Returns connected evidence paths rather than an unstructured bundle of similar documents.

Code Cell 06 | instructional infographic



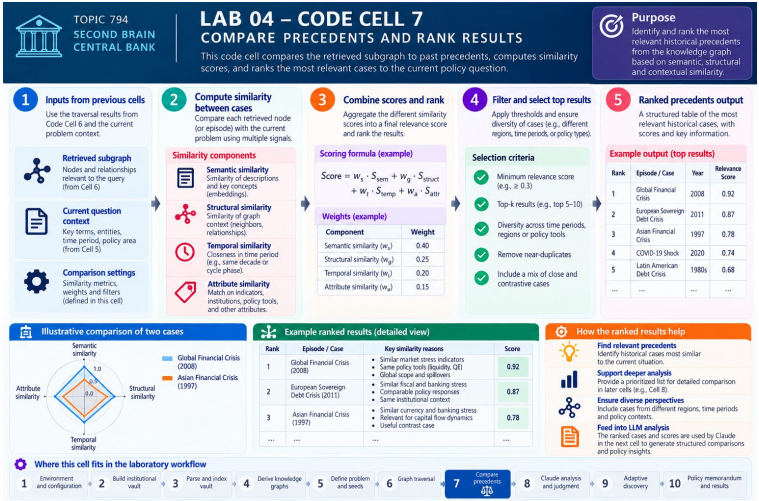
Code Cell 07 | Historical precedent assessment

METHOD AND INSTITUTIONAL DESIGN

- Compare three synthetic historical episodes: repo funding dislocation, FX funding squeeze and payments outage.
- Mechanism overlap, baseline precedent weights and traversal-discovery bonus produce a ranked evidence packet.

Key lesson Repo disruption is the strongest analogy; correlation is not proof of policy impact.

Code Cell 07 | instructional infographic



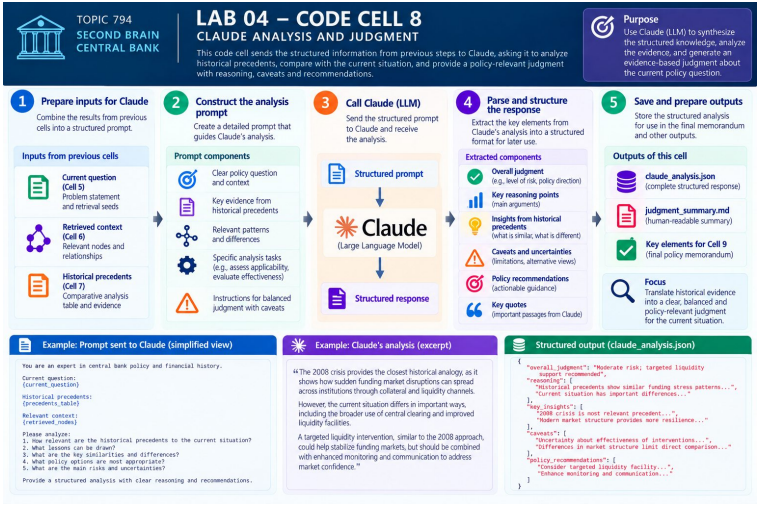
Code Cell 08 | Claude Haiku 5.5 judgment

METHOD AND INSTITUTIONAL DESIGN

- Pass the bounded evidence packet to Claude and request a policy assessment with cited source IDs and explicit caveats.
- Validate model output and source IDs; use a clearly labeled deterministic demonstration when live inference is unavailable.

Key lesson The LLM is an interpreter of bounded evidence—not the owner of institutional facts.

Code Cell 08 | instructional infographic



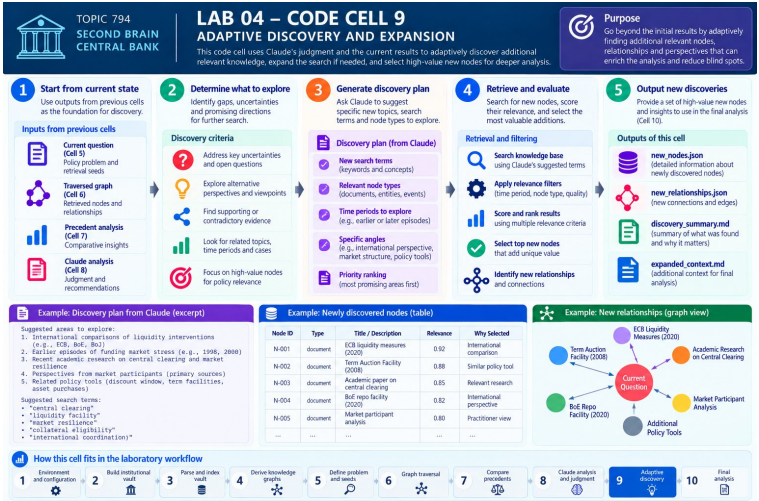
Code Cell 09 | Controlled adaptive discovery

METHOD AND INSTITUTIONAL DESIGN

- Permit the model to select at most one additional note from six predefined candidates; validate before admitting it.
- Enforce allowlisted IDs, temporal cutoff and access classification; record rejected and approved proposals.

Key lesson A model may influence what evidence is examined next without controlling the graph.

Code Cell 09 | instructional infographic



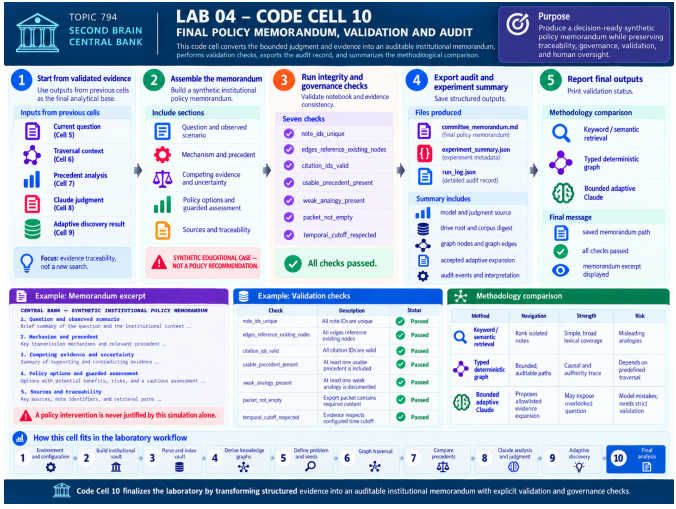
Code Cell 10 | Memorandum, evaluation and audit

METHOD AND INSTITUTIONAL DESIGN

- Format the existing judgment as a committee memorandum, run seven integrity checks and export auditable results.
- Writes committee_memorandum.md, experiment_summary.json and run_log.json; compares three methodologies.

Key lesson No new Claude call is made here: the final step packages, checks and documents the analysis.

Code Cell 10 | instructional infographic



Illustrations versus executable methodology

Reading these figures critically

- The infographics are pedagogical schematics, not execution logs.
- Some illustrations show broader possibilities than the narrow code actually implements.
- **Cells 7–10**: refer to deterministic scoring, bounded Claude assessment, single-note adaptive discovery and audited final export.

Implementation details are authoritative in the notebook.

Three methodological paradigms

Approach	Who chooses evidence?	Strength	Limitation
Semantic retrieval	Keyword or embedding ranker	Fast topical discovery	Weak structural and causal explanation
Governed traversal	Predefined graph rules	Traceable, auditable paths	Dependent on hand-crafted structure
Bounded adaptive discovery	LLM chooses a valid next node	Judgment guides evidence expansion	Must enforce access, cut-off, provenance

A precedent is not a causal conclusion

Symptom → mechanism → decision → instrument → outcome

- Historical coincidence does not identify causal intervention effects.
- Alternative explanations must survive graph retrieval.
- Temporal validity and institutional mandates affect transferability.
- The committee must separate observations, hypotheses and recommendations.

Governance and model boundaries

Machine-assisted tasks

- Navigate known nodes
- Compare mechanisms
- Summarize conflicting evidence
- Propose one permitted discovery
- Draft qualified judgment

Controls retained by the institution

- Stable IDs and provenance
- Scope and temporal filters
- Authorization and access
- Source validation
- Committee policy authority

What the laboratory demonstrates

1. A central bank can store its experience as inspectable, interlinked memory.
2. Typed graphs reveal relationships that ordinary text search obscures.
3. Historical learning needs both analogy and disanalogy.
4. LLM judgment is useful precisely when evidence access is governed.
5. Audit trails matter as much as fluent recommendations.

Closing thesis

From a knowledge repository to an institutional judgment system

Institutional intelligence

Persistent memory + relational structure + temporal discipline
+ governed traversal + bounded LLM judgment + audit

Selected readings and replication

- Karl Weick, *Sensemaking in Organizations* (1995).
- Douglass North, *Institutions, Institutional Change and Economic Performance* (1990).
- Judea Pearl, *Causality* (2009).
- David Lewis, *Counterfactuals* (1973).
- W3C, RDF 1.1 Concepts and Abstract Syntax (2014).
- Accompanying TOPIC 794 Colab notebook and LaTeX monograph.

Reproduce

Run cells 1–10 sequentially; synthetic vault and deterministic diagnostics work without a live LLM.